THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 7: **The Mathematics of Maybe: Introduction to Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{1}{2}$	favourable outcomes: 2, 3, 5 ⇒ 3 out of 6
2	**(b)** $\frac{1}{3}$	favourable outcomes: 1, 2 ⇒ 2 out of 6
3	**(a)** $\frac{3}{10}$	3 favourable out of 10 equally likely balls
4	**(d)** $\frac{2}{5}$	6 favourable out of 15 equally likely balls
5	**(b)** $\frac{2}{11}$	2 favourable out of 11 letters
6	**(b)** $\frac{4}{11}$	4 favourable out of 11 letters
7	**(c)** 0.4	$10/25 = 0.4$
8	**(b)** 36	multiply the number of outcomes of each experiment
9	**(a)** 0 and 1 (both included)	probability scale 0 to 1
10	**(c)** $\frac{1}{2}$	tosses are independent — the coin has no memory

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sample space $\{HH, HT, TH, TT\}$; favourable: HT, TH ⇒ $P = \frac{2}{4} = \frac{1}{2}$.	1+1
12	Primes: 2, 3, 5 ⇒ $\frac{3}{6} = \frac{1}{2}$.	1+1
13	Not red = blue: $\frac{6}{10} = \frac{3}{5}$.	1+1
14	Experimental $= \frac{8}{50} = 0.16$; theoretical $= \frac{1}{6} \approx 0.167$. They are close.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$ (1). At least two heads: HHH, HHT, HTH, THH — 4 outcomes (1). $P = \frac{4}{8} = \frac{1}{2}$ (1).	1+1+1
16	36 outcomes (1). (i) (2,6),(3,5),(4,4),(5,3),(6,2) ⇒ $\frac{5}{36}$ (1). (ii) (4,6),(5,5),(5,6),(6,4),(6,5),(6,6) ⇒ $\frac{6}{36} = \frac{1}{6}$ (1).	1+1+1
17	(i) $\frac{4}{52} = \frac{1}{13}$ (ii) $\frac{13}{52} = \frac{1}{4}$ (iii) $\frac{12}{52} = \frac{3}{13}$.	1+1+1
	OR: 10 balls. (i) $\frac{5}{10} = \frac{1}{2}$ (ii) $\frac{8}{10} = \frac{4}{5}$ (iii) 0 — impossible event.	1+1+1
18	$0.04 \times 1500 = 60$ bulbs (the estimate becomes better with a larger sample).	2+1
	OR: (i) $\frac{24}{120} = 0.2$ (1). (ii) $\frac{22+21+16}{120} = \frac{59}{120} \approx 0.49$ (1). Theoretical for 5 is $\frac{1}{6} \approx 0.167$; close but not equal (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{20}{50} = 0.4$ (1). (ii) apple 0.3, banana 0.2, grapes 0.1; sum $= 0.4 + 0.3 + 0.2 + 0.1 = 1$ ✓ (2). (iii) mango $0.4 \times 1500 = 600$; banana $0.2 \times 1500 = 300$ (1). (iv) Use a larger sample from different classes and grades (1).	1+2+1+1
	OR: (i) $6 \times 6 = 36$ (1). (ii) $\frac{6}{36} = \frac{1}{6}$ (1). (iii) $\frac{6}{36} = \frac{1}{6}$ (1). (iv) (1,1),(1,2),(2,1),(1,3),(3,1),(2,2) ⇒ $\frac{6}{36} = \frac{1}{6}$ (2).	1+1+1+2
20	(i) $\frac{1}{13}$ (ii) $\frac{1}{2}$ (iii) $\frac{3}{13}$ (iv) $\frac{26}{52} = \frac{1}{2}$ (v) $\frac{1}{52}$ (1 mark each).	1+1+1+1+1
	OR: (i) $\frac{5}{10}, \frac{3}{10}, \frac{2}{10}$; sum $= 1$ (2). (ii) $1 - \frac{3}{10} = \frac{7}{10}$ (1). (iii) 15 marbles, 7 green ⇒ $\frac{7}{15}$ (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 7: **The Mathematics of Maybe: Introduction to Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{1}{3}$	favourable outcomes: 5, 6 ⇒ 2 out of 6
2	**(b)** $\frac{5}{6}$	favourable outcomes: 1,2,3,4,5 ⇒ 5 out of 6
3	**(c)** $\frac{1}{3}$	4 favourable out of 12 equally likely balls
4	**(a)** $\frac{1}{3}$	5 favourable out of 15 equally likely balls
5	**(b)** $\frac{4}{11}$	4 favourable out of 11 letters
6	**(d)** 0.16	$8/50 = 0.16$
7	**(a)** 0.35	$28/80 = 0.35$
8	**(c)** 600	expected number $= P\times$ population
9	**(a)** 0	can never happen
10	**(d)** the Gambler's Fallacy	past outcomes do not affect the next fair toss

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$P = \frac{10}{50} = 0.2$ (1). Expected $= 0.2 \times 1500 = 300$ (1).	1+1
12	11 letters. (i) B occurs twice ⇒ $\frac{2}{11}$. (ii) Vowels O, A, I, I ⇒ $\frac{4}{11}$.	1+1
13	No. Each toss is independent; the coin has no memory, so $P(\text{tails})$ is still $\frac{1}{2}$. This mistaken belief is the Gambler's Fallacy.	1+1
14	(i) Certain (probability 1). (ii) Impossible (probability 0, or essentially so).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P = 0.35$; expected $= 0.35 \times 1200 = 420$ (2). A larger sample drawn from different classes represents the whole population better (1).	2+1
16	$\frac{1}{6}$ (1). Rolls are independent, so earlier results do not change it; the belief is the Gambler's Fallacy (2).	1+2
17	$P(\text{not } E) = 1 - \frac{5}{12} = \frac{7}{12}$ (2). Sum $= \frac{5}{12} + \frac{7}{12} = 1$ ✓ (1).	2+1
	OR: 11 letters. (i) $\frac{2}{11}$ (ii) $\frac{2}{11}$ (iii) vowels A, E, A, I ⇒ $\frac{4}{11}$.	1+1+1
18	$\frac{5}{500} = \frac{1}{100} = 0.01$.	2+1
	OR: (i) 6 doubles ⇒ $\frac{6}{36} = \frac{1}{6}$ (1). (ii) Primes 2,3,5,7,11 occur in $1+2+4+6+2 = 15$ ways ⇒ $\frac{15}{36} = \frac{5}{12}$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 0.15, 0.183, 0.158, 0.175, 0.2, 0.133 (2). (ii) $\frac{18+19+24}{120} = \frac{61}{120} \approx 0.51$ (1). (iii) All lie near 0.167 but differ (1). (iv) Each face would be near $\frac{1}{6}$, so about 1000 times (1).	2+1+1+1
	OR: (i) The mistaken belief that past random outcomes change the chances of the next one (1). (ii) $\frac{1}{2}$ (1). (iii) $\frac{1}{6}$ (1). (iv) Each toss or roll is an independent event; the coin and die have no memory (2).	1+1+1+2
20	(i) Certain (1). (ii) Impossible (0). (iii) Less likely ($\frac{1}{6}$). (iv) Even chance ($\frac{1}{2}$). (v) Certain (1). (1 mark each)	1+1+1+1+1
	OR: (i) 11 (1). (ii) $\frac{2}{11}$ (1). (iii) $\frac{2}{11}$ (1). (iv) Vowels O, A, I, I ⇒ $\frac{4}{11}$; consonants $\frac{7}{11}$; sum $= 1$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.